Shubhadeep Mandal

81602 63797 shubhadeepmandal11@gmail.com shubhadeep-mandal Seronic2001

Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Harpy: AI Competitive Programming Toolkit | *Python, MCP, C++, GitHub Actions*

- Built an MCP server that lets AI coding assistants (Cursor, Claude) compile, run and test C++ code locally.
- Added automatic stress testing against a brute-force solution and a sandboxed runner with time limits.
- Published it on PyPI and as a standalone binary, tested in CI across Python versions.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Exported the models to run fully in the browser and deployed them as a live demo.

BoardBuddy: Real-Time Collaborative Canvas | *Next.js, TypeScript, Convex, WebSockets*

- Built a real-time collaborative whiteboard with live cursors, drawing, shapes, sticky notes and undo/redo.
- Synced edits between users over WebSockets and stored boards in Convex.
- Checked every read and write on the server, so users can only access their own organisation's boards.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

Technical Skills

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL.
AI: PyTorch, Hugging Face, MCP, ONNX.
Web: Next.js, React, Node.js, Convex, Prisma, PostgreSQL.
Systems & Tools: Linux, Multithreading, Sockets, Docker, Git, GitHub Actions.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.